

Satellite Structural Inspection Interferometric Synthetic Aperture Radar (InSAR) Based

INDEX

Sr. No.	Description	Page No.
1.	Methodology	1
	1 Satellite SAR System with Resolution and Revisit Time	
	2 Retrieving Multi-Temporal SAR Imagery Over the Specified Period	
	3 Differential SAR Interferometry (D-InSAR) for Displacement Measurement	2
2.	Visualization	3
	1 Defining specific bridge location & parameters for satellite SAR based inspection	
	2 Calibration, georeferencing, orthorectification of SAR images (VH/VV)	4
	3 Results of SAR based bridge inspection, including visualizations	5
	3.1 VV Displacement From 23 Jan 2025 to 28 June 2025	6
	3.2 VH Displacement From 23 Jan 2025 to 28 June 2025	6
	3.3 SAR Based Analysis at Abutment 1 LHS	7
	3.4 SAR Based Analysis at Abutment 1 RHS	8
	3.9 SAR Based Analysis at Abutment 2 LHS	9
	3.10 SAR Based Analysis at Abutment 2 RHS	10
3.	Analysis	11
	1 Identify potential deformation, shifts or movement of the bridge structure	
	Conclusion	

METHODOLOGY

1. Satellite SAR System with Resolution and Revisit Time

To select a suitable SAR system, technical specifications such as spatial resolution, frequency band, polarization, and revisit time were considered. Sentinel-1, operating at C-band (5.405 GHz), was selected due to:

Parameter	Sentinel-1 C-SAR (IW Mode)
Frequency Band	C-band (~5.405 GHz)
Revisit Time	12 days (single satellite)
Spatial Resolution	5 m (range) × 20 m (azimuth)
Coverage Mode	Interferometric Wide Swath (IW)

2. Retrieving Multi-Temporal SAR Imagery Over the Specified Period

Multi-temporal **Sentinel-1 SLC IW mode** data were downloaded based on:

- **Area of Interest (AOI):** A polygon covering the bridge and surrounding terrain.
- **Date range:** Multiple acquisitions were collected spanning several months to capture temporal variations.
- **Image acquisition conditions:** All images were selected with the same relative orbit, pass direction (ascending or descending), and polarization (typically VV/VH) to ensure consistency for interferometric analysis.

S1A_IW_SLC__1SDV_20250111T130240_20250111T130310_057397_0710AB_813D
S1A_IW_SLC__1SDV_20250123T130239_20250123T130309_057572_07179E_5431
S1A_IW_SLC__1SDV_20250204T130238_20250204T130308_057747_071E86_3FE6
S1A_IW_SLC__1SDV_20250216T130238_20250216T130308_057922_072592_C312
S1A_IW_SLC__1SDV_20250228T130238_20250228T130307_058097_072CB4_7A14
S1A_IW_SLC__1SDV_20250312T130238_20250312T130308_058272_0733CA_217F
S1A_IW_SLC__1SDV_20250324T130238_20250324T130308_058447_073AB5_B445
S1A_IW_SLC__1SDV_20250405T130238_20250405T130308_058622_0741CE_2B52
S1A_IW_SLC__1SDV_20250417T130238_20250417T130308_058797_0748F4_3FFD
S1A_IW_SLC__1SDV_20250429T130239_20250429T130309_058972_07501E_08D8
S1A_IW_SLC__1SDV_20250511T130238_20250511T130308_059147_0756BA_DE13
S1A_IW_SLC__1SDV_20250523T130238_20250523T130308_059322_075CCB_14B2
S1A_IW_SLC__1SDV_20250604T130237_20250604T130307_059497_0762D9_3AB5
S1A_IW_SLC__1SDV_20250616T130237_20250616T130306_059672_0768D1_F57D
S1A_IW_SLC__1SDV_20250628T130236_20250628T130306_059847_076EE3_6BFD

3. Differential SAR Interferometry (D-InSAR) for Displacement Measurement

D-InSAR analysis was performed to measure millimetre-scale deformation over time:



VISUALIZATIONS

1. Defining specific bridge location & parameters for satellite SAR based inspection



Project & Bridge Details

State – Rajasthan

Bridge No. – 899/1

Bridge Chainage – 911+160

Project - Chittorgarh to Kota

Zone – RJ 06

Structure Type – VUP

Sar Parameters

Sensor – Sentinel-1 (C-Band)

Beam Mode – IW (Interferometric Wide Swath)

Path – 34

Frame – 519

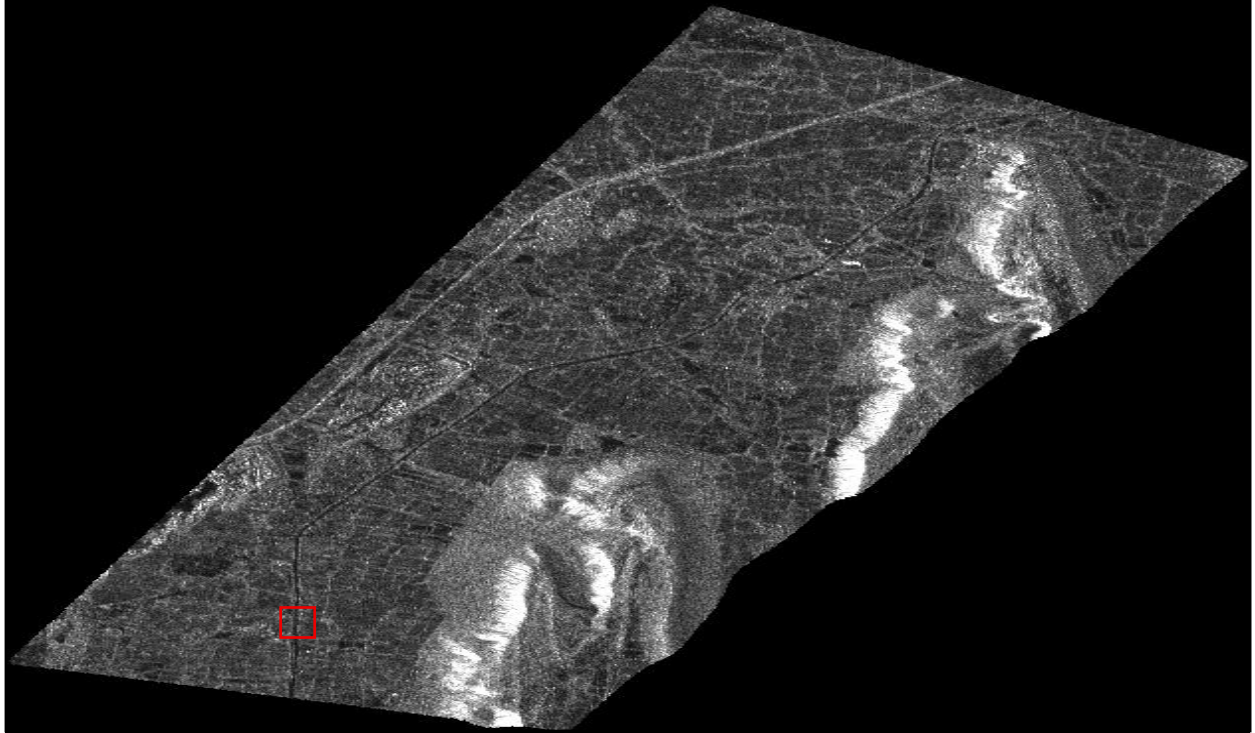
Flight Direction - DESCENDING

Polarization – VV+VH

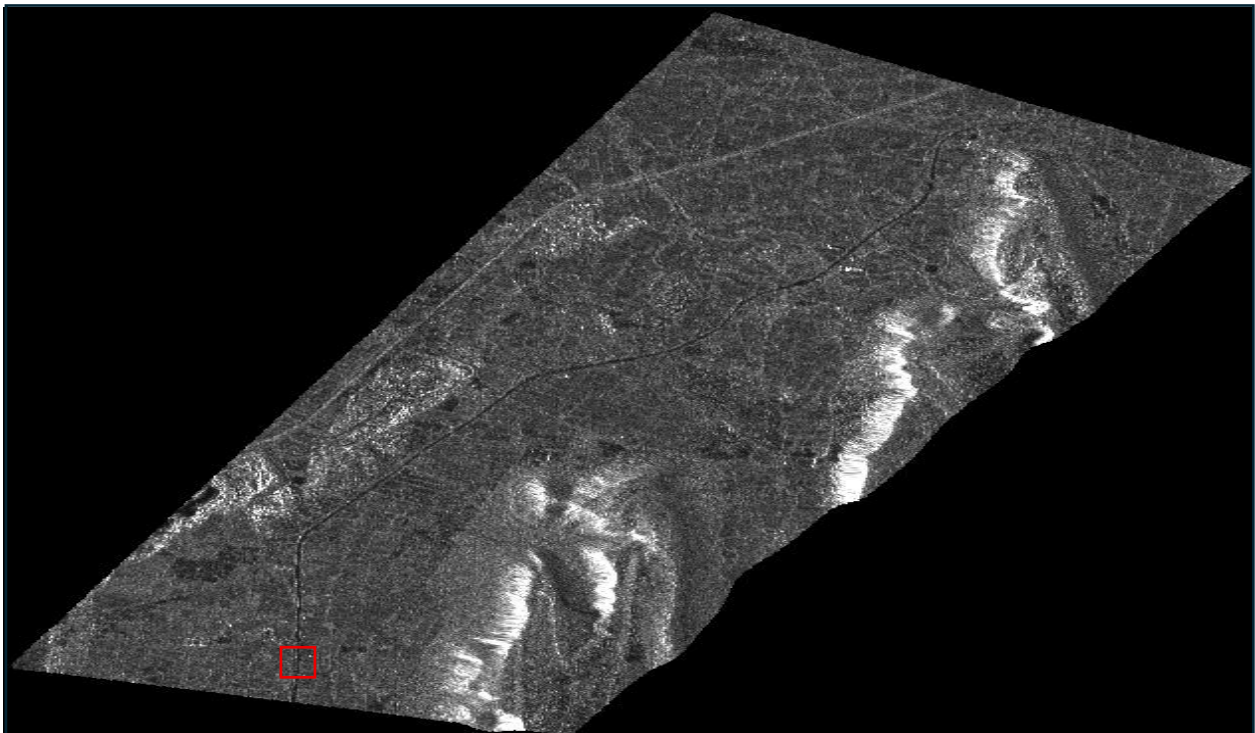
Absolute Orbit – 54706

2. Calibration, georeferencing, orthorectification of SAR images (VH/VV)

- Orthorectification of SAR image VH Polarization



- Orthorectification of SAR image VV Polarization

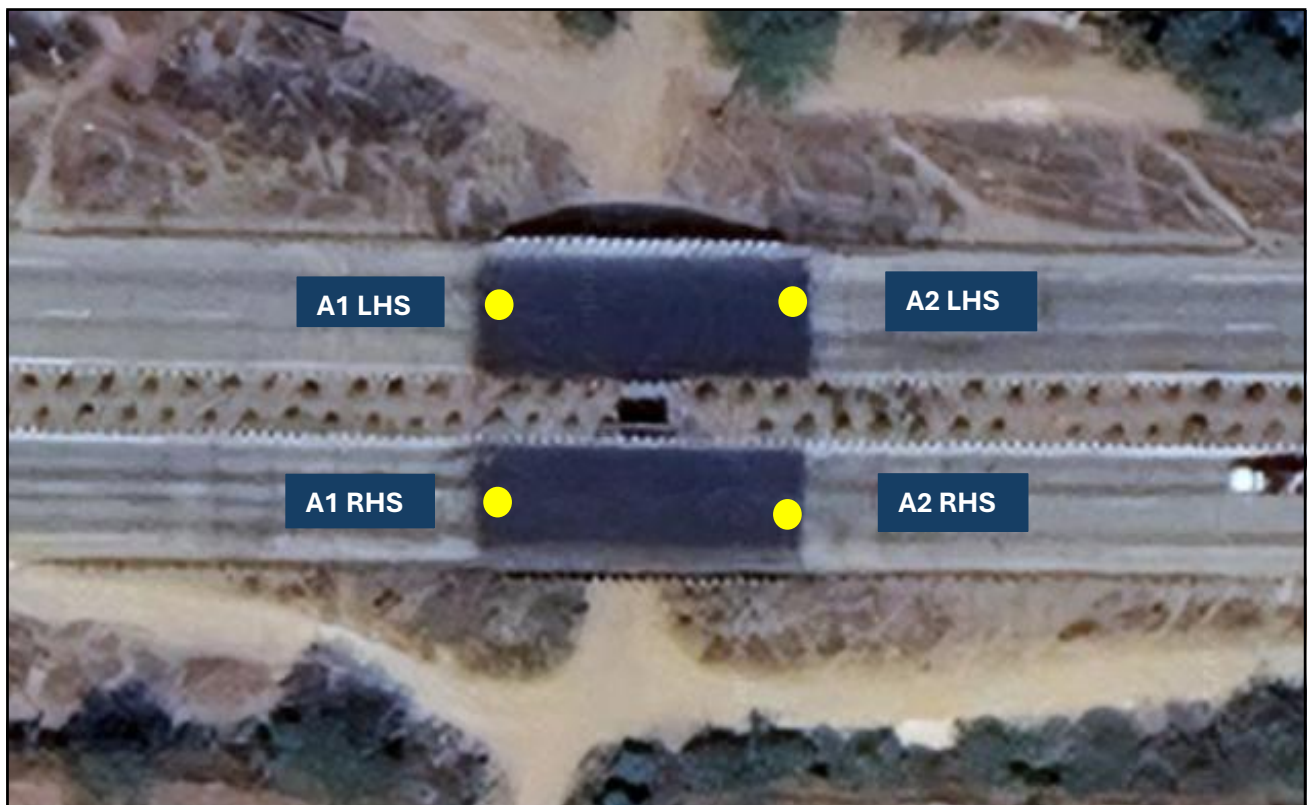


3. Results of SAR based bridge inspection, including visualizations

Bridge Components

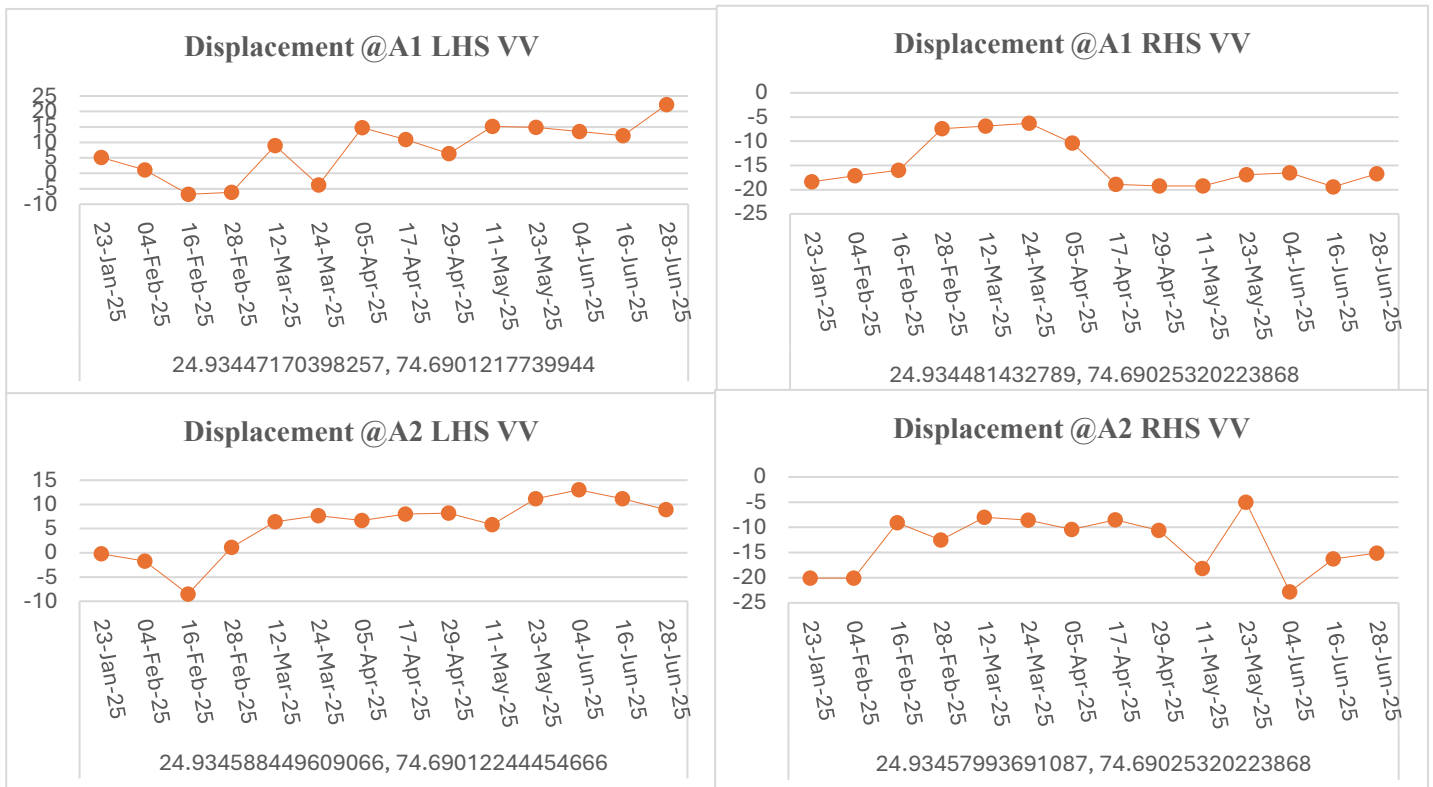
VUP – CH-911+160

Length – 22.78M

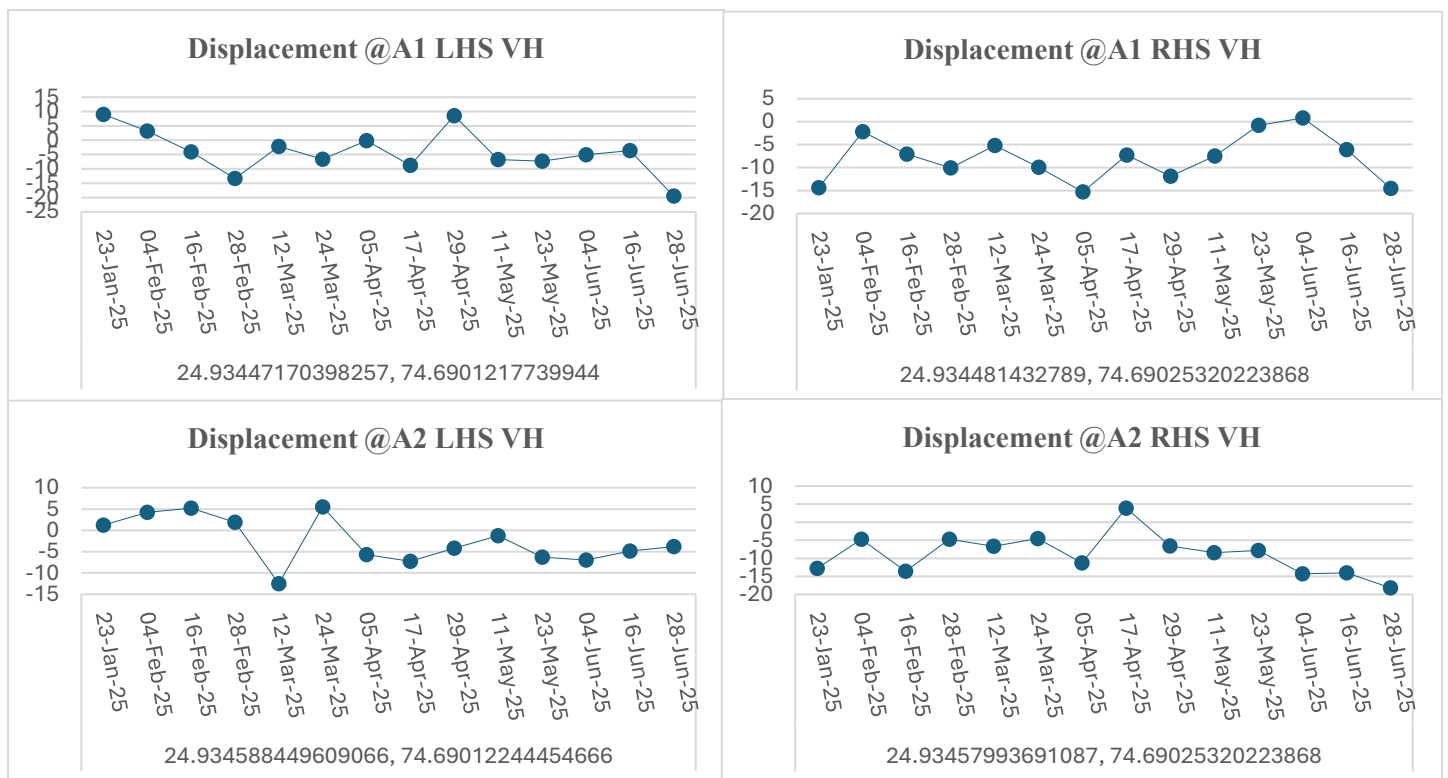


Scan Duration		From : 23 Jan 2025	To : 28 June 2025
Components & Abbreviation			
A1	Abutment 1		
A2	Abutment 2		
LHS	Left Hand Side		
RHS	Right Hand Side		

3.1 VV Displacement From 23 Jan 2025 To 28 June 2025



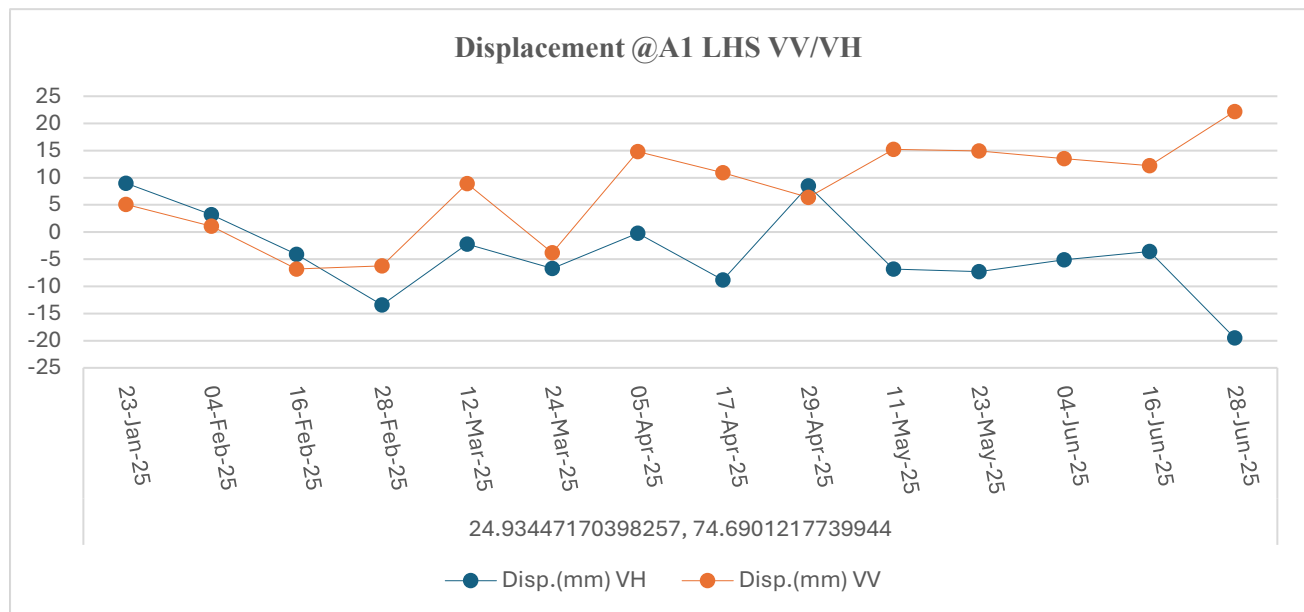
3.2 VH Displacement From 23 Jan 2025 To 28 June 2025



3.3 SAR Based Analysis at Abutment 1 LHS:



A1 LHS (24.93447170398257, 74.6901217739944)



Key Observations:

Displacement Magnitude Patterns:

- **Vertical Displacement (VV)** Vertical displacement varies approximately from –5 mm to +22 mm, exhibiting overall uplift-dominated behaviour with intermittent short-term settlement phases.
- **Horizontal Displacement (VH)** at A1 LHS ranges approximately from +10 mm to –20 mm, showing a predominantly negative trend after February 2025.

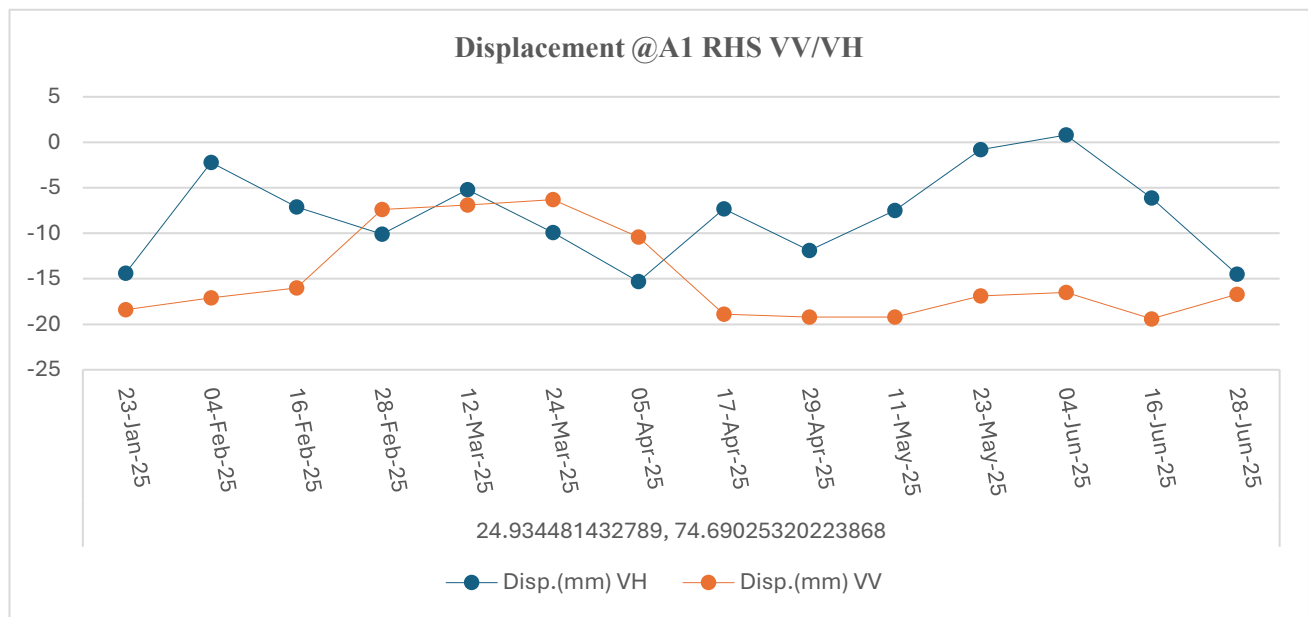
Critical Events & Anomalies:

- **February 2025:** VH transitions from positive to negative values, while VV remains low to moderate, suggesting initial lateral realignment or redistribution of forces.
- **June 2025:** VH reaches its maximum negative magnitude while VV peaks at 22 mm, indicating simultaneous significant lateral displacement and vertical uplift, characteristic of long-term deformation accumulation rather than transient effects.

3.4 SAR Based Analysis at Abutment 1 RHS:



A1 RHS (24.934481432789, 74.69025320223868)



Key Observations:

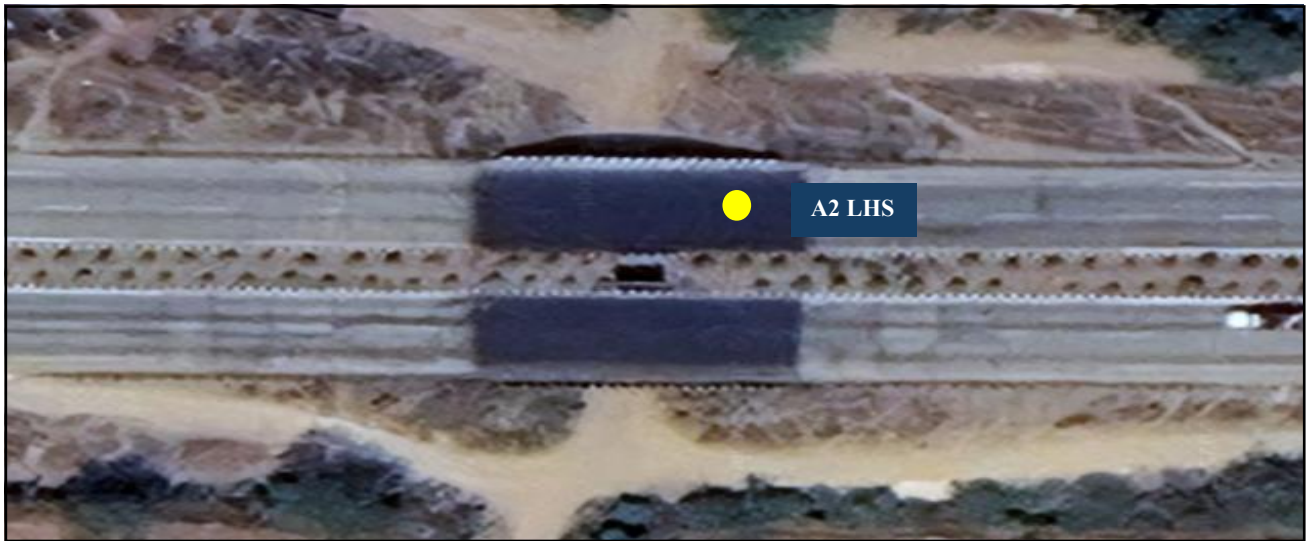
Displacement Magnitude Patterns:

- **Horizontal displacement (VH)** at A1 RHS remains entirely negative, ranging approximately from –15 mm to about –2 mm.
- **Vertical displacement (VV)** Vertical displacement ranges approximately from –6 mm to –20 mm, exhibiting a consistently negative trend throughout the monitoring period.

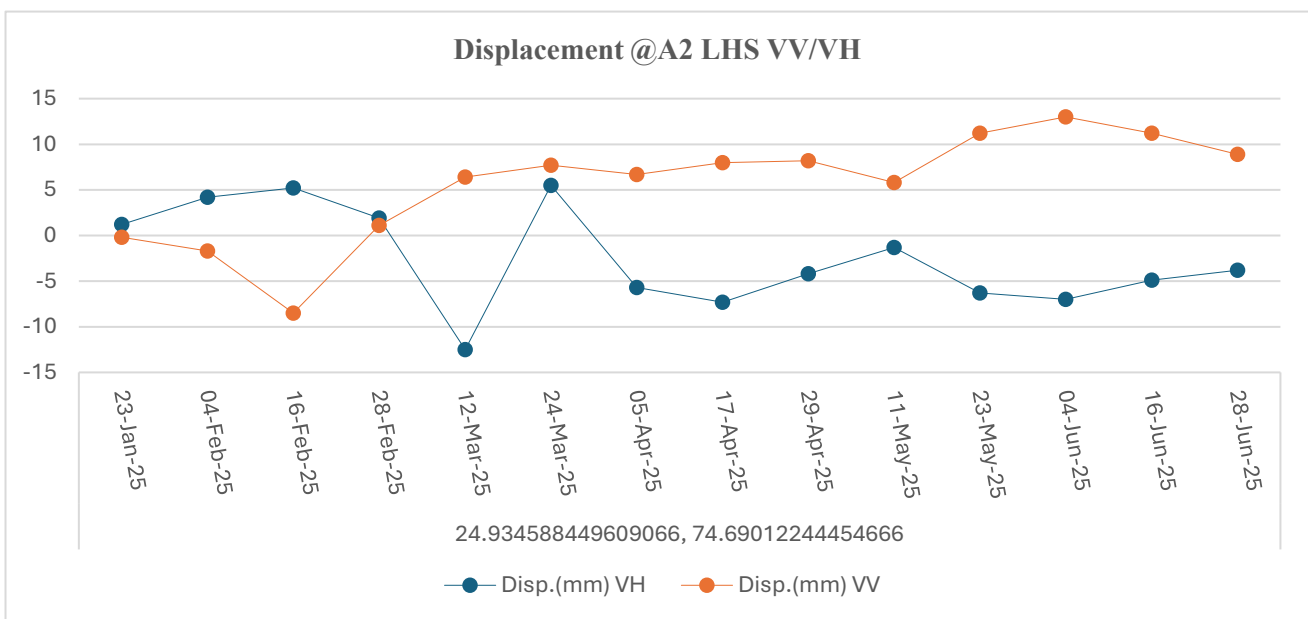
Critical Events & Anomalies:

- **February 2025:** Both VH and VV converge near –7 to –10 mm, indicating a brief phase of coupled lateral and vertical response, possibly linked to transient loading or thermal effects.
- **May 2025:** VH shows its least negative values (–2 to –1 mm) while VV remains significantly negative (–17 mm), suggesting lateral stabilization despite continued vertical settlement.
- **June 2025:** Both VH and VV trend back toward more negative values, indicating renewed lateral displacement accompanied by sustained settlement. characteristic of long-term deformation rather than isolated events.

3.5 SAR Based Analysis at Abutment 2 LHS:



A2 LHS (24.934588449609066, 74.69012244454666)



Key Observations:

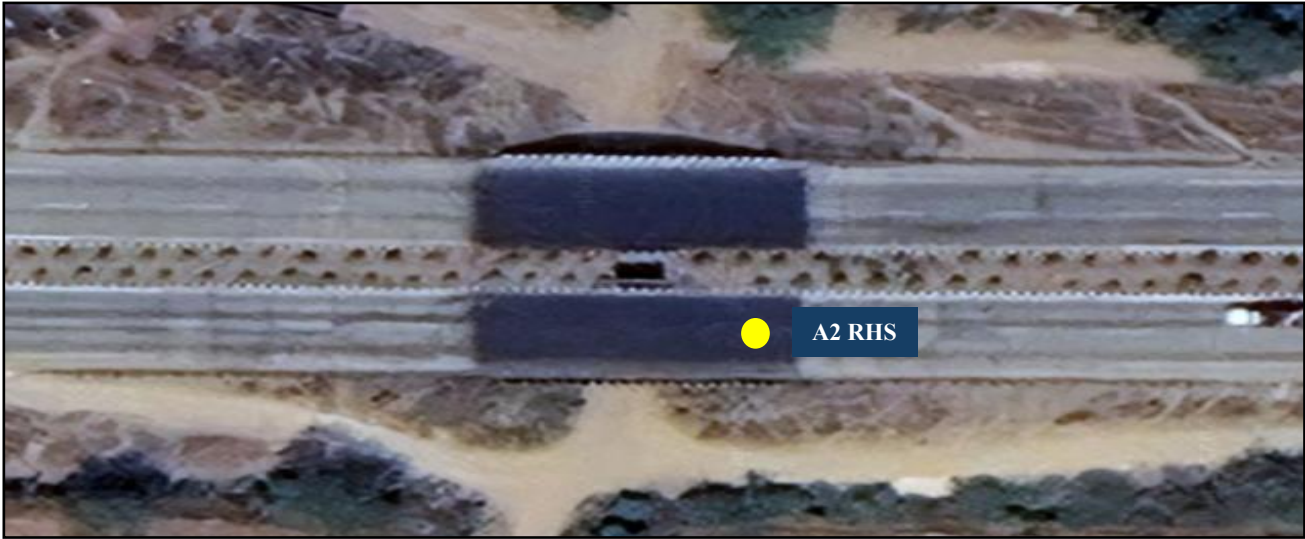
Displacement Magnitude Patterns:

- **Horizontal displacement (VH)** at A2 LHS ranges approximately from +5 mm to about -13 mm.
- **Vertical Displacement (VV)** transitions into a predominantly positive regime from March onward, reflecting sustained uplift or recovery with moderate fluctuations.

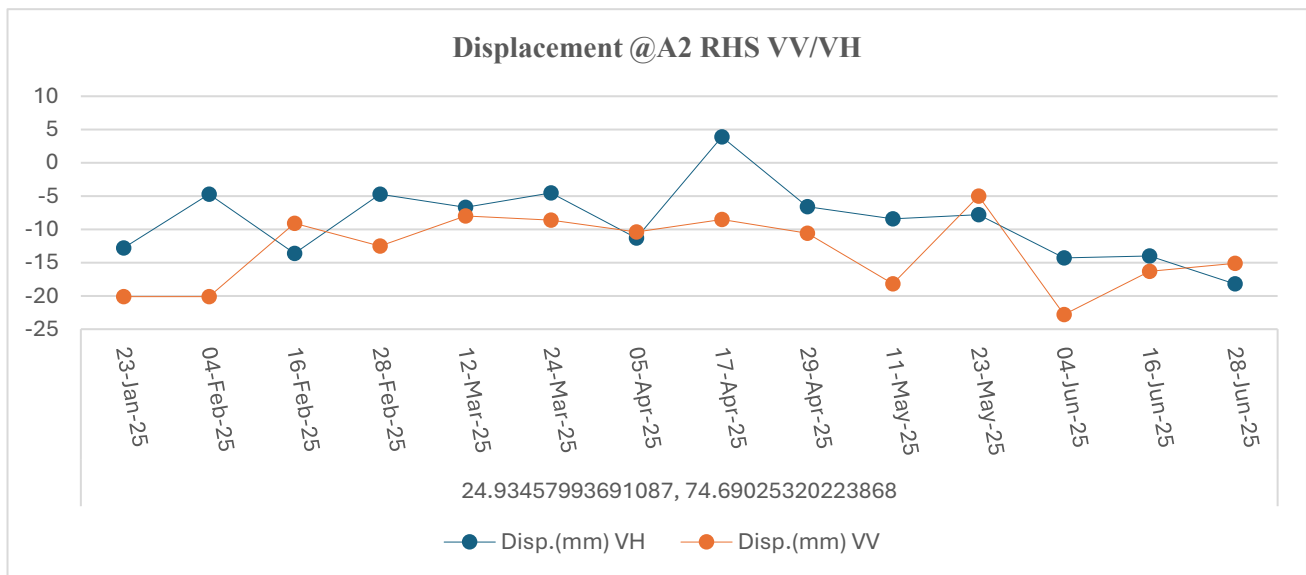
Critical Events & Anomalies:

- **March-April 2025:** VH partially recovers but remains negative, while VV stays consistently positive indicating vertical stabilization with continued lateral bias.
- **May 2025:** VV reaches its maximum uplift whereas VH remains moderately negative, pointing to dominant vertical movement with restrained lateral recovery.
- **June 2025:** VH stabilizes between -6 mm and -4 mm and VV shows a slight reduction from peak values, indicating overall stabilization with no abrupt deformation events.

3.6 SAR Based Analysis at Abutment 2 RHS:



A2 RHS (24.93457993691087, 74.69025320223868)



Key Observations:

Displacement Magnitude Patterns:

- **Vertical (VV) displacement** Vertical displacement varies approximately between –23 mm and –8 mm.
- **Horizontal (VH) displacement** Horizontal displacement at A2 RHS ranges approximately from +4 mm to –18 mm.

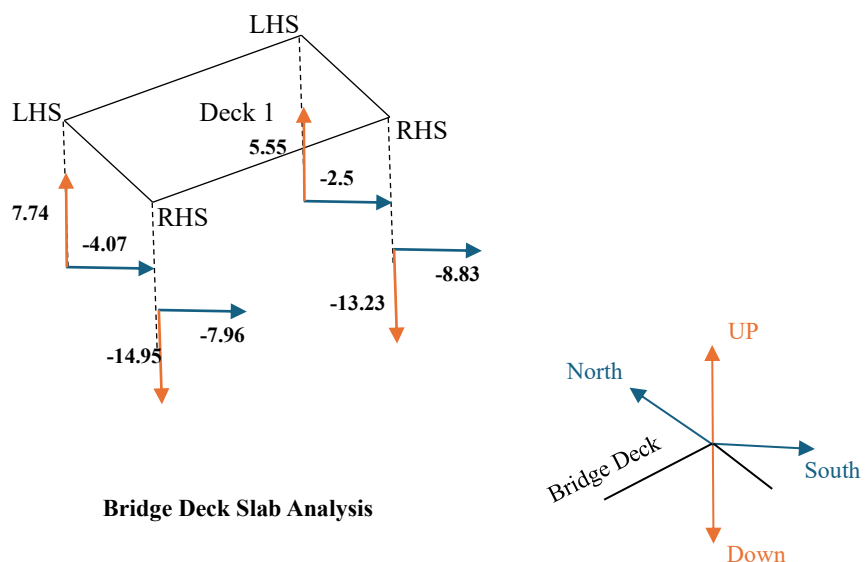
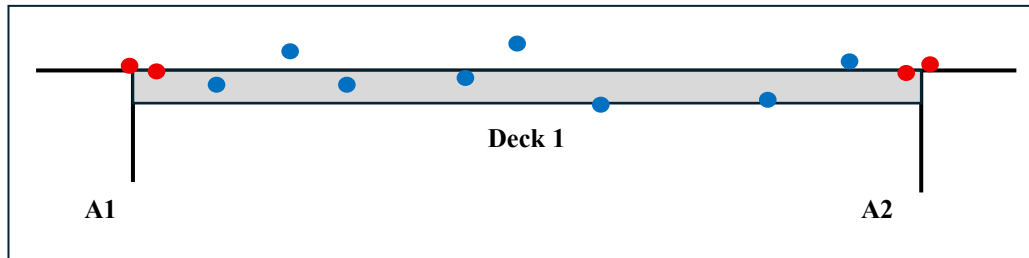
Critical Events & Anomalies:

- **January-February 2025:** VH fluctuates between –12 mm and –5 mm, while VV remains around –20 mm, suggesting early-stage lateral stability but ongoing vertical settlement.
- **May 2025:** VV reaches a maximum settlement, while VH again shifts more negative highlighting a critical period of combined lateral bias and vertical settlement.
- **June 2025:** Both VH and VV stabilize within negative ranges indicating post-settlement stabilization with no progressive deformation trend.

ANALYSIS

1. Identify potential deformation, shifts or movement of the bridge structure

- Points of Analysis (POA) of bridge (●)



Vertical Displacement Behaviour:

- The bridge deck slab exhibits mixed vertical behaviour, with uplift at the upper deck regions and settlement at the lower support regions, indicating a flexural response of the deck system. This vertical displacement pattern indicates hogging-sagging behaviour, with compression and tension zones developing across the deck depth.

Horizontal Displacement Behaviour:

- Horizontal displacements indicate moderate lateral movement predominantly towards the southward direction, as reflected by consistently negative values at most measurement points. The upper LHS region shows a horizontal displacement of approximately -4.07 mm, confirming asymmetric lateral response between LHS and RHS.

CONCLUSION:

- The bridge deck slab demonstrates a combined flexural and lateral response, with uplift at the deck level and notable settlement at the lower support regions. While displacement magnitudes remain within a moderate structural response range, the observed settlement pattern warrants continued monitoring, particularly at the lower central and RHS support locations, to detect any progression or long-term trends.